

Leaf-to-leaf paths and cycles in degree-critical graphs

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joint work with Jie Ma,

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Overview

- Consider distinct cycle lengths in some sparse graph family.
- Some of our proofs exploit a connection to an additive problem that might be of independent interest.

Degree k -critical graphs

Fact

Every n -vertex graph with at least $2n - 1$ edges contains a **proper** induced subgraph with minimum degree at least 3.

Proof: Greedily delete vertices of minimum degree.

Definition (Erdős, Faudree, Gyárfás, and Schelp, 1988)

For integer $k \geq 3$, a n -vertex graph with $(k - 1)n - \binom{k}{2} + 1$ edges and no **proper** induced subgraph with minimum degree at least k is called a *degree k -critical graph*.

Results on large cycle lengths in degree 3-critical graphs

- (Erdős, Faudree, Gyárfás, and Schelp, 1988) Every n -vertex degree 3-critical graph contains a cycle of length at least $\log n$.
- (Bollobás and Brightwell, 1989) Longest cycle length is at least $4 \log n + O(\log \log n)$, which is asymptotically best possible.

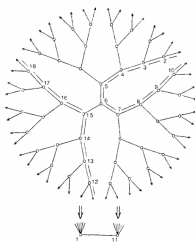


Figure: $4 \log n$ is asymptotically tight.

Results on large cycle lengths in degree 3-critical graphs

Conjecture 1 (Narins, Pokrovskiy, and Szabó 2017)

Every degree 3-critical graph on n vertices contains cycles of at least $3 \log n + O(1)$ distinct lengths.

Theorem (M., Braccio, Katsamaktsis, Malekshahian, Zhao, 2025+)

For $k \geq 3$, every degree k -critical graph on n vertices contains cycles of at least $\frac{\log n}{3 + \log k} + O(1)$ distinct lengths.

Results on small cycle lengths in degree 3-critical graphs

Question (Erdős, Faudree, Gyárfás, and Schelp)

Is it possible to find cycle lengths $3, 4, 5, \dots, N$ in any n -vertex degree 3-critical graph for some $N = N(n) \rightarrow \infty$ as $n \rightarrow \infty$?

- (Erdős et al.) Every n -vertex degree 3-critical graph contains a cycle of length 3, 4, and 5.

Theorem (Narins, Pokrovskiy, and Szabó 2017)

There are arbitrarily large degree 3-critical graphs with no 23-cycle.

Degree 3-critical graphs based on even 1-3 trees

- A tree is a 1-3 tree, if any vertex has degree 1 or 3.

For every 1-3 tree T , define G_T to be the graph obtained from T by adding a new edge xy , and connect $\{x, y\}$ to every leaf of T .

Proposition

G_T is degree 3-critical.

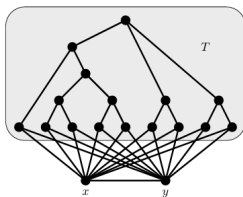


Figure: The degree 3-critical graph G_T .

Leaf-to-leaf path lengths in even 1-3 tree T

- Denote the set of all lengths of leaf-to-leaf paths in T by $\mathcal{P}_T$, and the set of all lengths of cycles in G by $\mathcal{C}_G$.

lemma

If T is a 1-3 tree such that any root-leaf path has even length, then

- $\mathcal{C}_{G_T} \cap (2\mathbb{N} + 1) = \mathcal{P}_T + 3$.
- $\mathcal{C}_{G_T} \cap (2\mathbb{N}) \subseteq (\mathcal{P}_T + 2) \cup (\mathcal{P}_T + \mathcal{P}_T + 4)$.

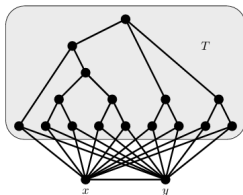


Figure: The degree 3-critical graph G_T .

Conjecture for small leaf-to-leaf path lengths

- (Narins, Pokrovskiy, and Szabó) Large 1-3 trees do not necessarily contain a leaf-to-leaf path of length 20.

Conjecture 2 (Szabó et al.)

There exist a constant $\alpha > 0$ and a function $N = N(n)$ tending to infinity as $n \rightarrow \infty$ such that every 1-3 tree of order n contains at least αN distinct leaf-to-leaf path lengths between 0 and N .

Theorem (M., Braccio, Katsamaktsis, Malekshahian, Zhao, 2025+)

Let $c = (2 - \log 10 / \log 13)^{-1} \approx 0.9073$. For all $N \geq 1$ and all even $n \geq N$, there exists an n -vertex 1-3 tree with $O(N^c)$ distinct leaf-to-leaf path lengths between 0 and N .

Connection to a problem in additive combinatorics

The constant $c = (2 - \log 10 / \log 13)^{-1} \approx 0.9073$ is related to the following construction in additive combinatorics.

- Let $A = \{1, 2, 5, 7\}$, then $A - A = [-6, 6]$ and $|A + A| = 10$.

Fact

There exist sufficient large sets $U, V \subseteq \mathbb{Z}$ such that $\log |U + V| / \log |U - V| = \log 10 / \log 13$.

- U, V are translations of $\left\{ \sum_{0 \leq i \leq k-1} a_i 13^i : a_i \in A \right\}$.

Lower bounds for small leaf-to-leaf path lengths

We also have a poly-sublinear lower bound by considering distinct leaf-to-leaf paths sharing a common endpoint.

Theorem (M., Braccio, Katsamaktsis, Malekshahian, Zhao, 2025+)

For N sufficiently large, let T be a sufficiently large tree containing no vertex of degree 2, then $|\mathcal{P}_T \cap [0, N]| = \Omega(N^{2/3})$ where these paths share a common endpoint. The exponent $2/3$ is tight.

The optimal c for $|\mathcal{P}_T \cap [0, N]| = \Omega(N^c)$ is related to the solution of $\min\{\log |U + V| / \log |U - V|\}$.

Conjecture for distinct cycle lengths

Paralleling to distinct cycle lengths in degree 3-critical graphs, Szabó et al. also conjectured distinct leaf-to-leaf path lengths in 1–3 trees.

Conjecture 3 (Szabó et al.)

Every 1–3 tree T of order n has leaf-to-leaf paths of at least $\log(n+2) - 1$ distinct lengths.

Theorem (M., Braccio, Katsamakis, Malekshahian, Zhao, 2025+)

Let T be a tree with maximum degree $\Delta \geq 3$ and ℓ leaves. Then T has at least $\log_{\Delta-1}((\Delta-2)\ell)$ distinct leaf-to-leaf path lengths, tight for infinitely many n .

- When T is a 1-3 tree, $|T| = 2\ell - 2$.

Ideas

We will sketch the proof ideas for the following two results.

Theorem (M., Braccio, Katsamaktsis, Malekshahian, Zhao, 2025+)

Every degree k -critical graph on n vertices contains cycles of at least $\frac{\log n}{3+\log k} + O(1)$ distinct lengths.

Theorem (M., Braccio, Katsamaktsis, Malekshahian, Zhao, 2025+)

Let $c = (2 - \log 10 / \log 13)^{-1} \approx 0.9073$. For all $N \geq 1$ and all even $n \geq N$, there exists an n -vertex 1-3 tree with $O(N^c)$ distinct leaf-to-leaf path lengths between 0 and N .

Ideas for distinct cycle lengths in degree 3-critical graphs

Theorem (M., Braccio, Katsamakis, Malekshahian, Zhao, 2025++)

Every degree k -critical graph on n vertices contains cycles of at least $\frac{\log n}{3 + \log k} + O(1)$ distinct lengths.

Let $x_1, x_2, \dots, x_n$ be an ordering of $V(G)$. Define

- forward degree $d^+(x_i) = |\{x_j \in N_G(x_i) : i < j\}|$,
- backward degree $d^-(x_i) = |\{x_j \in N_G(x_i) : i > j\}|$.

Proposition

Every degree k -critical graph G is quipped with an ordering $\mathcal{X} = x_1, x_2, \dots, x_n$ of $V(G)$, such that

$$d^+(x_i) = \begin{cases} k, & i = 1 \\ k - 1, & 2 \leq i \leq n - k \\ n - i, & i \geq n - k + 1 \end{cases}$$

Proof ideas

Every graph equipped with such an ordering $\mathcal{X}$ contains sufficient distinct cycle lengths.

Sketch:

- A partial order $\preceq$ on $V(G)$: $u \preceq v \iff$ there is an "forward" path from u to v under $\mathcal{X}$.
- By Dilworth's theorem, there is either a large chain or antichain under $\preceq$.
- Either a large chain or a large antichain $\implies$ many distinct cycle lengths.

Case: Long chain

- A long chain $\iff$ a long forward path in G .
- A *vine* structure based on P (introduced by Bondy, 1981) contains a cycle of length in $(|P|, 2|P|]$.

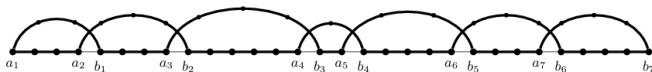


Figure: A vine starting at a_1 contains a cycle (bold).

- If there is a chain of size t , then $|\mathcal{C}_G| \geq \log t - 1$.

Case: Large antichain

- For an anti-chain L , there is a rooted tree with leaf set L , such that every root-leaf path is forward.
- Find subtrees with constant depth.

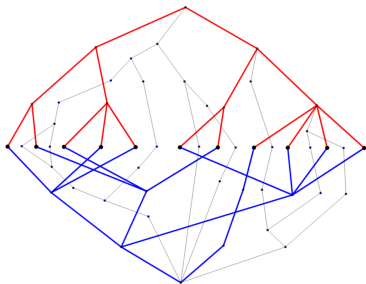


Figure: A structure implied by an anti-chain.

Ideas for sparse leaf-to-leaf path length in fixed interval

Theorem (M., Braccio, Katsamaktsis, Malekshahian, Zhao, 2025+)

Let $c = (2 - \log 10 / \log 13)^{-1} \approx 0.9073$. For all $N \geq 1$ and all even $n \geq N$, there exists an n -vertex 1-3 tree with $O(N^c)$ distinct leaf-to-leaf path lengths between 0 and N .

Construction: starting with a path P , a binary tree of depth a_i is attached to each inner vertex of P .

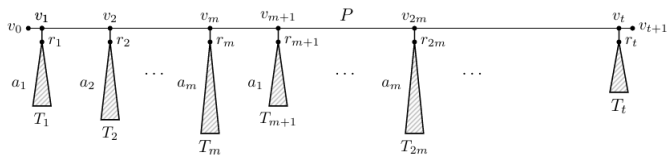
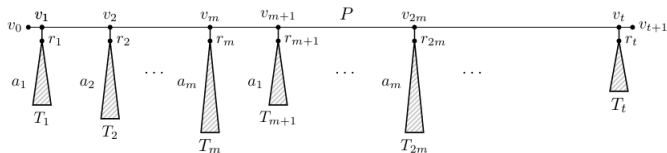


Figure: A special 1-3 tree $T = T((a_1, \dots, a_t))$.

Construction of the depth sequence (a_i)



Sketch:

- Select $U, V \subseteq \mathbb{Z}$ with $U - V = [13^k]$, $|U + V| = 10^k$, and $U + V \subseteq U - V$.
- Let $M = \lfloor (13^2/10)^k \rfloor \approx N$, we show that $|\mathcal{P}_T \cap [M]| = O(13^k)$.
- There exists sequence $(a_i) \subseteq U + V$ such that $a_i + a_j + j - i \in 2 \cdot (U + V) + 13^k \cdot \mathbb{Z}$ for any $1 \leq i < j \leq t$.

Open problems

Problem D

Determine the supremum $c \in [0, 1]$ for which the following holds: for all N and all sufficiently large even n , every n -vertex 1–3 tree contains leaf-to-leaf paths of $\Omega(N^c)$ distinct lengths between 0 and N .

Problem E

Is there a function $C(n)$ tending to infinity such that every degree 3-critical graph on n vertices contains cycles of all **even** lengths $4, 6, 8, \dots, 2C(n)$?

Thanks.